Philanthropy during the MATS extension program, which made this project possible.

Author Contributions

MP discovered the attention-based constrained belief updating algorithm in the simplex, and performed the bulk of the experiments with mentorship from ASS. PMR developed the mathematical theory together with MP and ASS. ASS supervised the project, and DF provided project management. MP, PMR, and ASS wrote the manuscript, with helpful guidance from DF. MP, PMR, and ASS performed analysis, and established the correspondence between transformer behavior and theoretical predictions.

Impact Statement

This paper presents work whose goal is to advance the interpretability of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

Bricken, T., Templeton, A., Batson, J., Chen, B., Jermyn, A., Conerly, T., Turner, N., Anil, C., Denison, C., Askell, A., Lasenby, R., Wu, Y., Kravec, S., Schiefer, N., Maxwell, T., Joseph, N., Hatfield-Dodds, Z., Tamkin, A., Nguyen, K., McLean, B., Burke, J. E., Hume, T., Carter, S., Henighan, T., and Olah, C. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features/index.html.

Conmy, A., Mavor-Parker, A. N., Lynch, A., Heimersheim, S., and Garriga-Alonso, A. Towards automated circuit discovery for mechanistic interpretability, 2023. URL https://arxiv.org/abs/2304.14997.

Crutchfield, J. P. The calculi of emergence: Computation, dynamics, and induction. *Physica D*, 75:11–54, 1994.

Crutchfield, J. P. Between order and chaos. *Nature Physics*, 8(1):17–24, 2012.

Cunningham, H., Ewart, A., Riggs, L., Huben, R., and Sharkey, L. Sparse autoencoders find highly interpretable features in language models, 2023. URL https://arxiv.org/abs/2309.08600.

Elhage, N., Nanda, N., Olsson, C., Henighan, T., Joseph, N., Mann, B., Askell, A., Bai, Y., Chen, A., Conerly, T., DasSarma, N., Drain, D., Ganguli, D., Hatfield-Dodds, Z., Hernandez, D., Jones, A., Kernion, J., Lovitt, L., Ndousse, K., Amodei, D., Brown, T., Clark, J., Kaplan, J., McCandlish, S., and Olah, C. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. URL https://transformer-circuits.pub/2021/framework/index.html.

Elhage, N., Hume, T., Olsson, C., Schiefer, N., Henighan, T., Kravec, S., Hatfield-Dodds, Z., Lasenby, R., Drain, D., Chen, C., Grosse, R., McCandlish, S., Kaplan, J., Amodei, D., Wattenberg, M., and Olah, C. Toy models of superposition. *Transformer Circuits Thread*, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.

Engels, J., Michaud, E. J., Liao, I., Gurnee, W., and Tegmark, M. Not all language model features are linear. *arXiv preprint arXiv:2405.14860*, 2024.

Kaelbling, L. P., Littman, M. L., and Cassandra, A. R. Planning and acting in partially observable stochastic domains. *Artificial intelligence*, 101(1-2):99–134, 1998.

Kingma, D. P. and Ba, J. Adam: A method for stochastic optimization, 2017. URL https://arxiv.org/abs/1412.6980.

Marks, S., Rager, C., Michaud, E. J., Belinkov, Y., Bau, D., and Mueller, A. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models, 2024. URL https://arxiv.org/abs/2403.19647.

Marzen, S. E. and Crutchfield, J. P. Nearly maximally predictive features and their dimensions. *Physical Review E*, 95(5), May 2017. ISSN 2470-0053. doi: 10.1103/physreve.95.051301. URL http://dx.doi.org/10.1103/PhysRevE.95.051301.

Nanda, N., Chan, L., Lieberum, T., Smith, J., and Steinhardt, J. Progress measures for grokking via mechanistic interpretability, 2023. URL https://arxiv.org/abs/2301.05217.

Park, K., Choe, Y. J., and Veitch, V. The linear representation hypothesis and the geometry of large language models, 2024. URL https://arxiv.org/abs/2311.03658.

Pepper, K. RNNs represent belief state geometry in their hidden states. https://apartresearch.com, June 2024. Research submission to the Computational Mechanics Hackathon research sprint co-hosted by Apart, PIBBSS, and Simplex.

Riechers, P. M. and Crutchfield, J. P. Beyond the spectral theorem: Decomposing arbitrary functions of nondiagonalizable operators. *AIP Advances*, 8:065305, 2018a.

Riechers, P. M. and Crutchfield, J. P. Spectral simplicity of apparent complexity, Part I: The nondiagonalizable metadynamics of prediction. *Chaos*, 28:033115, 2018b. doi: 10.1063/1.4985199.

Shai, A. S., Marzen, S. E., Teixeira, L., Oldenziel, A. G., and Riechers, P. M. Transformers represent belief state geometry in their residual stream. *accepted to Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. URL https://arxiv.org/abs/2405.15943.

Shalizi, C. R. and Crutchfield, J. P. Computational mechanics: Pattern and prediction, structure and simplicity. *J. Stat. Phys.*, 104:817–879, 2001a.

Shalizi, C. R. and Crutchfield, J. P. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of statistical physics*, 104:817–879, 2001b.

Templeton, A., Conerly, T., Marcus, J., Lindsey, J., Bricken, T., Chen, B., Pearce, A., Citro, C., Ameisen, E., Jones, A., Cunningham, H., Turner, N. L., McDougall, C., MacDiarmid, M., Freeman, C. D., Sumers, T. R., Rees, E., Batson, J., Jermyn, A., Carter, S., Olah, C., and Henighan, T. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*, 2024. URL https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html.

Upper, D. R. *Theory and algorithms for hidden Markov models and generalized hidden Markov models*. University of California, Berkeley, 1997.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A., Kaiser, L., and Polosukhin, I. Attention is all you need. *Advances in Neural Information Processing Systems*, 2017.

Wang, K., Variengien, A., Conmy, A., Shlegeris, B., and Steinhardt, J. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small, 2022. URL https://arxiv.org/abs/2211.00593.

A. Mathematical Details of HMMs and Belief State Geometry

In this work we created training data from a class of Hidden Markov Models (HMMs) called Mess3. The HMMs have three hidden states $\mathcal{S} = \{1, 2, 3\}$ and emit from a vocabulary of three tokens $\mathcal{Z} = \{0, 1, 2\}$.

The HMMs in this class are parameterized by α and x, with dependent quantities $\beta = (1 - \alpha)/2$ and $y = 1 - 2x$.

The labeled transition matrices define the probability of moving to state j (indexing columns) and emitting the token on the label, z, conditioned on being in state i (indexing rows), $P(s_j, z|s_i)$ and are:

$$T^{(0)} = \begin{bmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{bmatrix} \tag{21}$$

$$T^{(1)} = \begin{bmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{bmatrix} \tag{22}$$

$$T^{(2)} = \begin{bmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{bmatrix} \tag{23}$$

Note that even though the dynamics amongst the emissions are infinite-Markov order, the dynamics amongst the hidden states are Markov, with a transition matrix given by marginalizing out the token emissions: $T = \sum_{z \in \mathcal{Z}} T^{(z)}$.

Since Mess3 has non-zero row sums for each labeled transition matrix, we can also define a conditional transition matrix, $T^{|z}$, with elements $T^{|z}_{i,j} = P(s_j|z, s_i)$, which is given by normalizing each labeled transition matrix such that every row sums to 1.

A.1. Full belief updates

An important part of the work presented here is about how an optimal observer of token emissions from the HMM would update their beliefs over which of the hidden states the HMM is in, given a token sequence. If the observer is in a belief state given by a probability distribution $\boldsymbol{\eta}$ (a row vector) over the hidden states of the data-generating process, then the update rule for the new belief state $\boldsymbol{\eta}'$ given that the observer sees a new token z is:

$$\boldsymbol{\eta}' = \frac{\boldsymbol{\eta} T^{(z)}}{\boldsymbol{\eta} T^{(z)} \mathbf{1}} \tag{24}$$

where $\mathbf{1}$ is a column vector of ones of appropriate dimension, with the denominator ensuring proper normalization of the updated belief state. In general, starting from the initial belief state $\boldsymbol{\eta}_\varnothing$, we can find the belief state after observing a sequence of tokens $z_0, z_1, \ldots, z_N$:

$$\vec{r}_{\text{full}}^{(z_{1:\text{d}})} = \frac{\boldsymbol{\eta}_\varnothing T^{(z_0)} T^{(z_1)} \ldots T^{(z_N)}}{\boldsymbol{\eta}_\varnothing T^{(z_0)} T^{(z_1)} \ldots T^{(z_N)} \mathbf{1}} \ . \tag{25}$$

For stationary processes, the optimal initial belief state is given by the stationary distribution $\boldsymbol{\eta}_\varnothing = \boldsymbol{\pi}$ over hidden states of the HMM (the left-eigenvector of the transition matrix $T = \sum_z T^{(z)}$ associated with the eigenvalue of 1).

The beliefs have a geometry associated with them, called the belief-state geometry. The belief-state geometry is given by plotting the belief distribution over the HMM's hidden states induced from each possible sequence of tokens as a point in the probability simplex over these hidden states.

A.2. Constrained belief updates

Incorporating past contributions to belief updates in parallel, as the attention mechanism suggests, we instead obtain

$$\vec{r}_1^{(z_{1:\text{d}})} = \boldsymbol{\pi} + \sum_{n=0}^{\text{d}-1} \Big(\frac{\boldsymbol{\pi} T^{(z_{\text{d}-n})} T^n}{\boldsymbol{\pi} T^{(z_{\text{d}-n})} \mathbf{1}} - \boldsymbol{\pi} \Big) \tag{26}$$